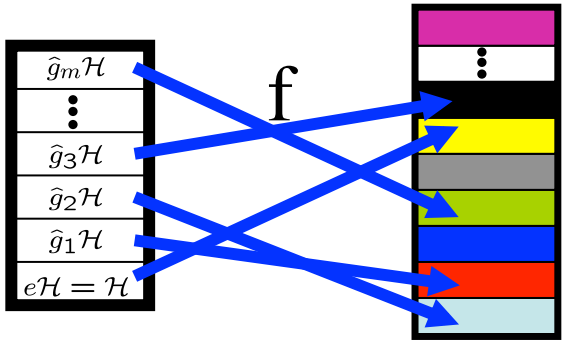
The Hidden Subgroup Conjugacy Problem and Beyond

Dave Bacon, University of Washington

dabacon@gmail.com / http://www.cs.washington.edu/homes/dabacon/qw/

Objective

• New quantum algorithms for hidden subgroup problems
• Understanding the relationship between the hidden subgroup conjugacy problem and the hidden subgroup problem
• Efficient quantum circuits for Clebsch-Gordan transforms over finite groups
• Understanding and exploiting the role of symmetry in quantum algorithms



$$\rho_{[\mathcal{H}]}^{\otimes m} = \bigoplus_{\mu_1,\ldots,\mu_m} \left(I_D \otimes \bigoplus_{\mu} \left(R_{[\mathcal{H}]} \otimes I_{d_\mu} \right) \right)$$

Objective Approach

• Focus on the double role symmetry takes on in efficient quantum algorithms
  1st role makes query complexity low
  2nd role makes efficient circuits possible
• Work with "more symmetric" hidden subgroup conjugacy problem instead of hidden subgroup problem
• Use of techniques from previous approaches including subgroup adapted basis, generalized Kitaev circuits, and "pretty good measurements"
• Exploiting the well understood structure of Clebsch-Gordan transforms for finite groups

Status

• New start

• In the prior year + a little:
• Developed optimal measurement technique for nonabelian hidden subgroup problem *
• Efficient quantum algorithm for the Heisenberg group *
• Efficient quantum circuits for the Schur and Clebsch-Gordan transform **

* with Andrew Childs (Caltech), Wim van Dam (UCSB)
** with Ike Chuang (MIT) and Aram Harrow (Bristol)



The Hidden Subgroup Conjugacy Problem and Beyond

Dave Bacon, University of Washington



• Progress on last years objectives
• New start

• Research plan for next 12 months
• Are the hidden subgroup conjugacy problem and the hidden subgroup problem equivalent? For which groups are they equivalent?
• Explicit understanding of the Clebsch-Gordan symmetry involved in the dihedral and symmetric group hidden subgroup conjugacy problem.
• Begin study of the construction of efficient circuits for finite group Clebsch-Gordan transforms.
• Identification of optimal measurement for single copy hidden subgroup problem

• Long term objectives (demonstrations)

• Algorithms for non-abelian hidden subgroup problems. (Take heed of lower bounds, but the goal is upper bounds.)
• Efficient quantum algorithms for graph isomorphism and certain k-shortest vector in a lattice problems
• Construction of efficient quantum algorithms for use in problem with symmetries (beyond the hidden subgroup problem: exploitation of symmetries of simulations of quantum systems)

• Very long term goals: New quantum algorithms. Wash our brains of what we've learned and come up with totally new approaches for efficient quantum algorithms.

Newbie

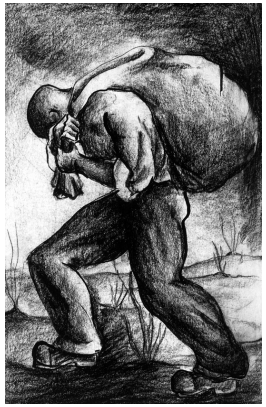
1996:

Quantum Computation and NP-Completeness

Dave Bacon
Summer Undergraduate Research Fellowship Paper
Advisors: Nicholas Cerf and Chris Adami
California Institute of Technology


Caltech ,


Berkeley ,


Caltech ,

and


Santa Fe Institute

2005:

From optimal measurement to efficient quantum algorithms for the hidden subgroup problem over semidirect product groups

Dave Bacon*
dabacon@santafe.edu

Andrew M. Childs†
amchilds@caltech.edu

Wim van Dam‡
vandam@cs.ucsb.edu

Results from Past Year + A Little

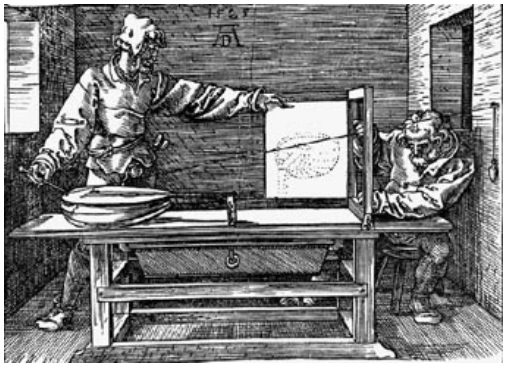
[Bacon, Childs, and van Dam 05]
Optimal PGM for dihedral HSP

[Bacon, Childs, and van Dam 05]
Efficient algorithm for the Heisenberg HSP

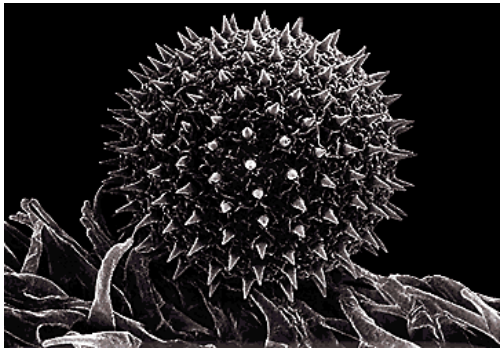
[Bacon, Harrow, Chuang 04-06]
Efficient transform for Schur transform and Clebsch-Gordan transform for U(d)

FORWARD LOOKING

Outline



Perspective
- Highlight role of symmetry in successes
- Why can we solve the Heisenberg HSP?



The Hidden Subgroup Conjugacy Problem
- Importance
- Symmetry
- Open problems



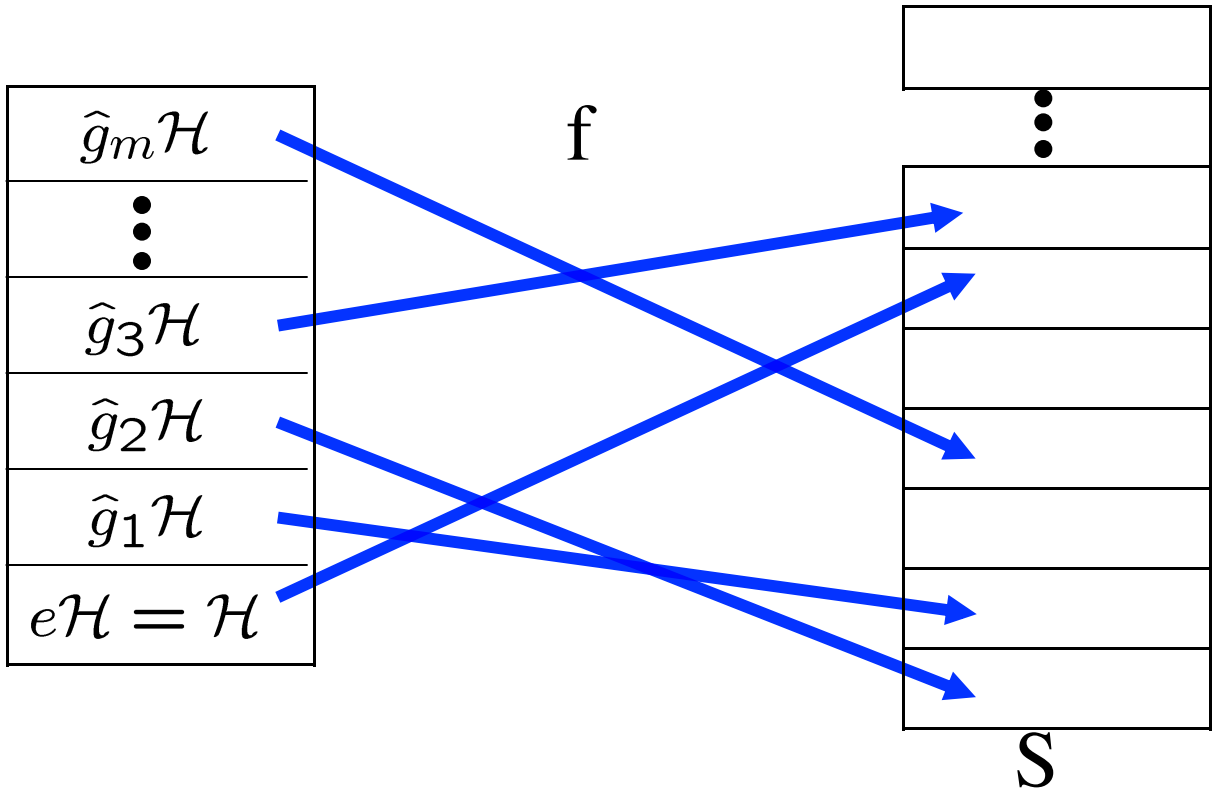
The Future
- Transforms?
- Algorithms?

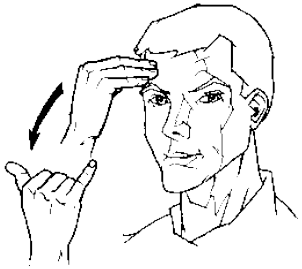
The Hidden Subgroup Problem

Setup: Fix a group $\mathcal{G}$ (known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ (unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$. We say that f hides $\mathcal{H}$.

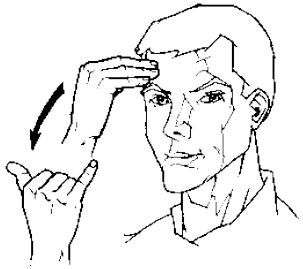
Problem: Find a generating set for $\mathcal{H}$ ("identify $\mathcal{H}$").

Efficient: If runs in time polylog($|\mathcal{G}|$)





Why the Fuss?



Solved Problems (Shor 1994)

efficient algorithm for HSP over $\mathbb{Z}_N$ → efficient algorithm for factoring whole numbers

efficient algorithm for HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$ → efficient algorithm for the discrete logarithm problem

Open Problems

efficient algorithm for HSP over $\mathcal{S}_n \wr \mathbb{Z}_2$ (or $\mathcal{S}_{2n}$) → efficient algorithm for the graph isomorphism problem

efficient "standard" quantum algorithm for HSP over dihedral group $\mathcal{D}_N$ → efficient algorithm for the poly(*n*) unique shortest vector in a lattice problem

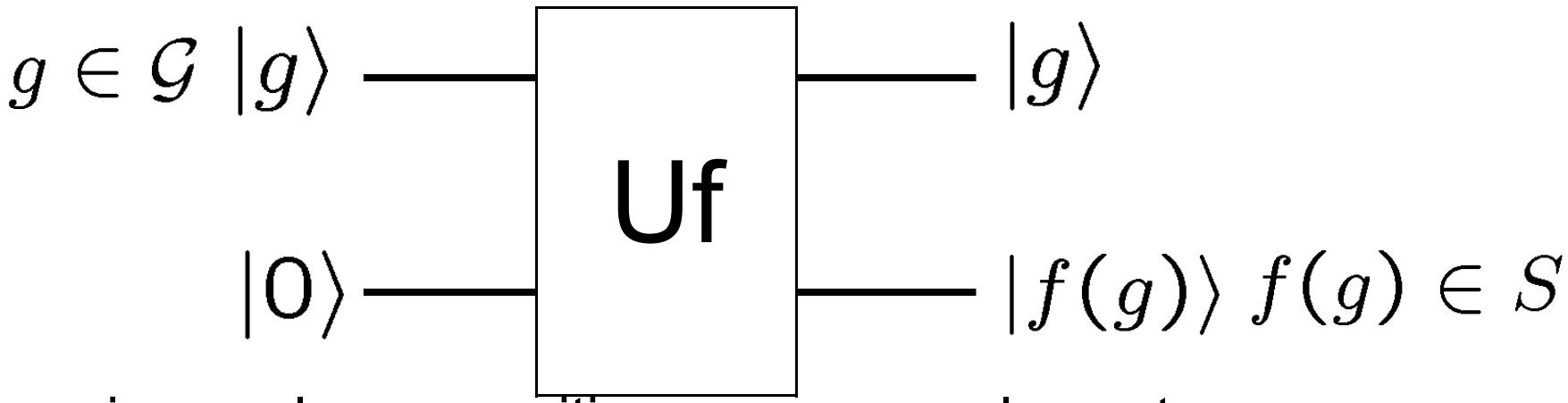
Hidden Subgroup Successes

- Abelian groups [Shor 94; Kitaev 95]
- $\mathbb{Z}_2^n \rtimes \mathbb{Z}_2$ [Püschel, Rötteler, Beth 98]
- Normal subgroups [Hallgren, Russell, Ta-Shma 00]
- "Amost Abelian" groups [GSVV 01]
- $\mathbb{Z}_{p^k}^n \rtimes \mathbb{Z}_2$ [FIMMS 02]
- "Weyl Hamiltonian" groups [Gavinsky 04]
- Some q-hedral groups [MRRS 04]
- Some $\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$ groups [Inui, Le Gall 04]
- Heisenberg-type groups [Bacon, Childs, and van Dam 05]

Why **THESE** successes?

Standard Query Model

Oracle:



Query in equal superposition over group elements

$$\frac{1}{\sqrt{|\mathcal{G}|}} \sum_{g \in \mathcal{G}} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|\mathcal{G}|}} |g\rangle|f(g)\rangle$$

constant and distinct on different left cosets of

$$|g\mathcal{H}\rangle = \frac{1}{\sqrt{|\mathcal{H}|}} \sum_{h \in \mathcal{H}} |gh\rangle$$ random left coset state

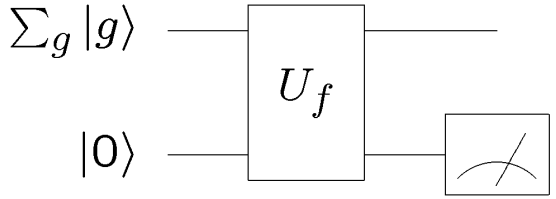
$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}|$$ hidden subgroup state

$\hat{\mathcal{G}}$:=set of coset representatives

Start Simple

$$\mathcal{G} = \mathbb{Z}_2 = \{e, r\}$$

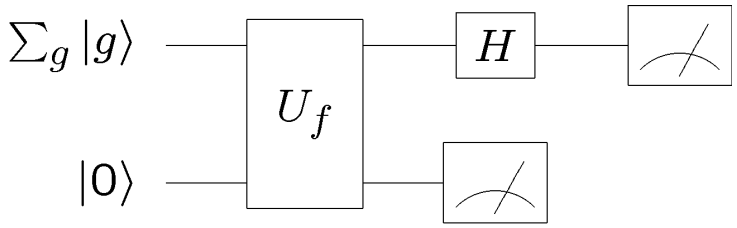
	e	r
e	e	r
r	r	e



Two possible hidden subgroups

$$\mathcal{H}_t = \{e\} \text{ or } \mathcal{H}_2 = \{e, r\}$$

$\mathcal{H}_t$	$\mathcal{H}_2$
50% $\|0\rangle\langle 0\|$	$\|+\rangle\langle +\|$
50% $\|1\rangle\langle 1\|$	$\|\pm\rangle = \frac{1}{\sqrt{2}}(\|0\rangle \pm \|1\rangle)$
$\rho_t = \frac{1}{2}I$	$\rho_2 = \|+\rangle\langle +\|$



Symmetry, Role I

I give you a symmetric probability distribution

$$T\vec{p} = \vec{p}$$

how does this help you?

I give you a symmetric quantum density matrix

$$T\rho T^\dagger = \rho$$

how does this help you?

T, ρ can be simultaneously diagonalized!

$$\rho_t = \frac{1}{2} I$$

$$\rho_2 = |+\rangle\langle +|$$

$$X\rho_h X^\dagger = \rho_h \Rightarrow$$

diagonalize in $|\pm\rangle$ basis

Symmetries As a Guide

The more symmetries we can identify, the more likely we are able to distinguish between different quantum states.

Example: Heisenberg HSP [Bacon, Childs, and van Dam 05]

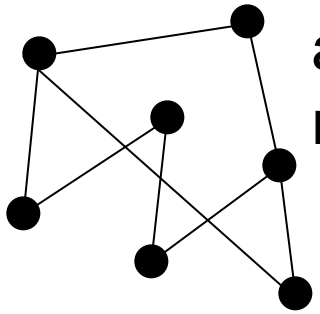
After two standard queries, algorithm queries over multiple solutions to certain quadratic equations:

$$v = b_1 x_1 + b_2 x_2$$

$$w = \frac{b_1(1-b_1)}{2} x_1 + b_1 y_1 + \frac{b_2(1-b_2)}{2} x_2 + b_2 y_2$$

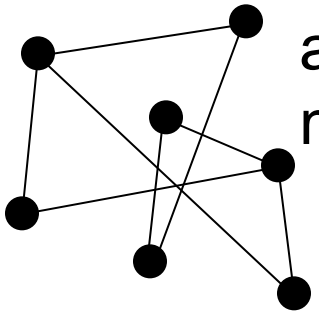
i.e. the symmetry which is essential to solving the problem is the symmetry group for the roots of these equations

Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



adjacency matrix A_2

$G_2 = (V_2, E_2)$

Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the vertices V_2 such that the edges are respected, $p(E_1) = E_2$?

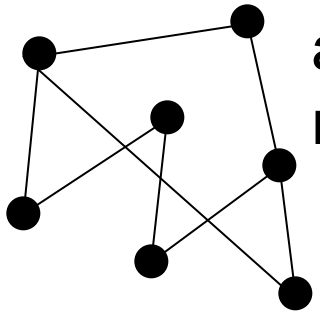
Mapping to the HSP:

$$f : \mathcal{S}_{2n} \rightarrow \mathbb{Z}_{n(2n-1)}$$

$$f(\pi) = \pi(A_1 \oplus A_2)$$

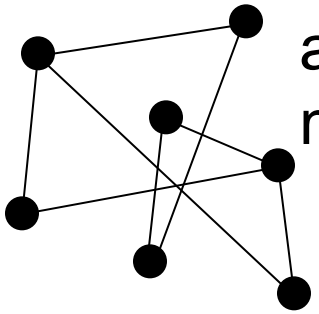
If graphs are not isomorphic, hidden subgroup will not mix the graphs. If they are isomorphic, they will mix the graphs.

Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



adjacency matrix A_2

$G_2 = (V_2, E_2)$

Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the vertices V_2 such that the edges are respected, $p(E_1) = E_2$?

Mapping to the HSP:

$$f : \mathcal{S}_n \wr \mathcal{S}_2 \rightarrow \mathbb{Z}_{n(2n-1)}$$

$$f((\pi_1, \pi_2, \tau)) = \tau(\pi_1(A_1) \oplus \pi_2(A_2))$$

Notice now we are only worried if hidden subgroup has a nontrivial τ

Conjugate Subgroups

Two subgroups, $\mathcal{H}_1$ and $\mathcal{H}_2$ are conjugate if there exists an element of the group $g \in \mathcal{G}$ such that

$$\mathcal{H}_1 = \{ghg^{-1}, h \in \mathcal{H}_2\}$$

This is an equivalence relation among the subgroups of $\mathcal{G}$

$$\mathcal{H}_i \equiv_{\mathcal{G}} \mathcal{H}_j$$

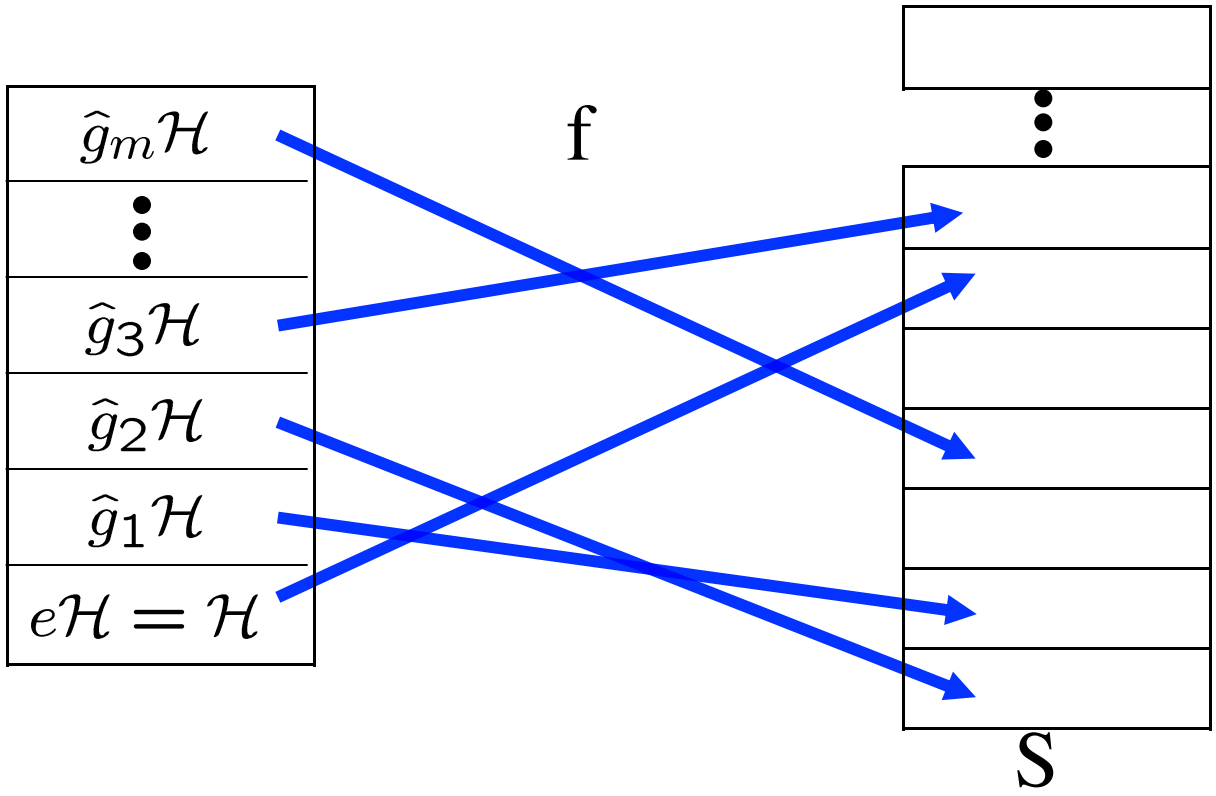
ir: voy, vas, va, vamos, van

Hidden Subgroup Conjugacy Problem

Setup: Fix a group $\mathcal{G}$ known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$ We say that f hides $\mathcal{H}$

Problem: identify the conjugate subgroup class of $\mathcal{H}$

Efficient: If runs in time polylog($|\mathcal{G}|$)



Relationships

How do the hidden subgroup problem (HSP) and the hidden subgroup conjugacy problem (HSCP) differ?

1. If group is abelian, HSCP=HSP

All subgroups are only conjugate to themselves

$$g\mathcal{H}g^{-1} = \mathcal{H}$$

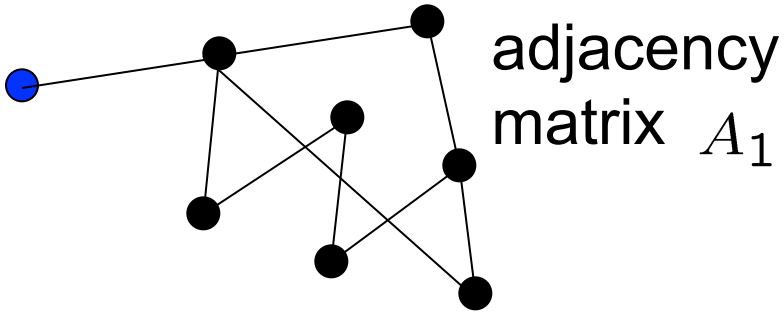
2. If subgroups are normal, HSCP=HSP

All subgroups are only conjugate to themselves

$$g\mathcal{H}g^{-1} = \mathcal{H}$$

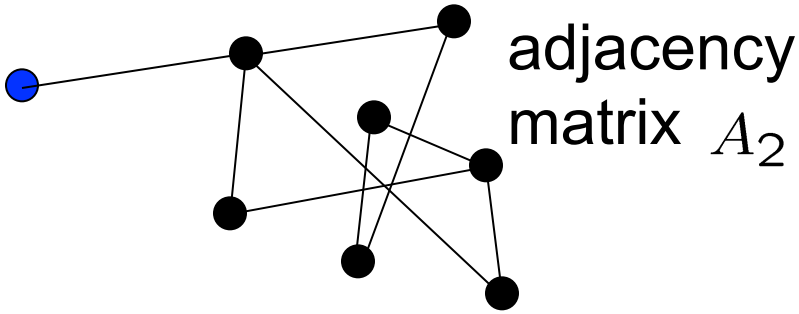
These are two cases where we know quantum computers can efficiently solve the HSP. Coincidence?

Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



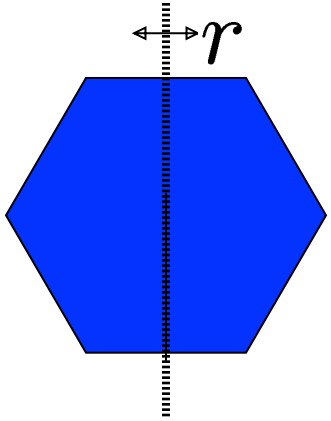
adjacency matrix A_2

$G_2 = (V_2, E_2)$

If we want to find the isomorphism, we can attach extra vertices and edges to detect whether permutation of these vertices is in isomoprhism group.

HSCP $\rightarrow$ solve graph isomorphism and identify the isomorphism

Dihedral Group



s = rotation

r = reflection

Dihedral group $\mathcal{D}_N$

group of symmetries of a *N* sided regular polygon

$$\mathcal{D}_N = \{r^t s^k, t \in \mathbb{Z}_2, k \in \mathbb{Z}_N\}$$

$$r^2 = s^N = rsrs = e$$

Subgroups of $\mathcal{D}_N$

cyclic subgroups:

$$\mathcal{C}_{N/j} = \{e, s^j, s^{2j}, \ldots, s^{-j}\}$$

$$j \in \mathbb{Z}_N \quad j \in \mathbb{Z}_N \quad j|N$$

dihedral subgroups:

$$\mathcal{D}_{N/j,l} = \{e, s^j, \ldots, s^{-j}, rs^l, rs^{j+l}, \ldots, rs^{-j+l}\}$$

$$j \in \mathbb{Z}_N \quad j \in \mathbb{Z}_N \quad j|N \quad l \in \mathbb{Z}_N$$

Dihedral HSP

Theorem [Ettinger, Hoyer 98]

Sufficient to solve Dihedral HSP when subgroups are limited to

$$\mathcal{H}_l = \{e, rs^l\} \quad l \in \mathbb{Z}_N \quad \text{or} \quad \mathcal{T} = \{e\}$$

if N is odd, then the $\mathcal{H}_l = \{e, rs^l\}$ subgroups are all conjugate to each other:

$$s^{-k}\mathcal{H}_l s^k = \{e, rs^{l+2k}\}$$

solving the Dihedral HSCP does not immediately solve the Dihedral HSP…

Modification



Modify the function hiding the subgroup:

$$f : \mathcal{D}_N \to S$$
$$g : \mathcal{D}_N \to \{+1, -1\}$$
$$g(r^t s^k) = (-1)^k$$

left cosets:

$$\mathcal{H}_l = \{e, rs^l\}$$
$$s\mathcal{H}_l = \{s, rs^{l-1}\}$$
$$s^2\mathcal{H}_l = \{s, rs^{l-2}\}$$
$$\vdots$$

If l is even, then (f, g) is constant and distinct on left cosets of $\mathcal{H}_l$

If l is odd, then (f, g) is constant and distinct on left cosets of $\mathcal{T} = \{e\}$

Solving HSCP for new problem thus allows us to distinguish between odd and even subgroups

[Bacon, Moore unpublished 05]

Modification

Modify the function hiding the subgroup:

$$f : \mathcal{D}_N \to S$$
$$g_1 : \mathcal{D}_N \to \{+1, -1\}$$
$$g_2 : \mathcal{D}_N \to \{+1, -1\}$$

left cosets:

$$\mathcal{H}_l = \{e, rs^l\}$$
$$s\mathcal{H}_l = \{s, rs^{l-1}\}$$
$$s^2\mathcal{H}_l = \{s, rs^{l-2}\}$$
$$\vdots$$

Using more functions, we can determine 2nd least significant bit of l

Generalizing this allows us to use the Dihedral HSCP to solve the Dihedral HSP (they are polynomial time equivalent.)

[Bacon, Moore unpublished 05]

see also [Kuperberg 03]

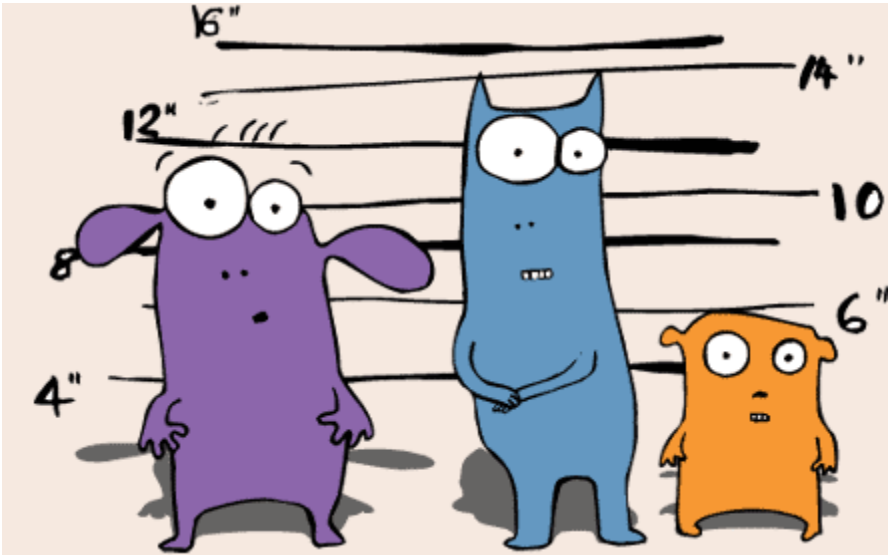
HSP and HSCP

More generally, we can ask if this method can be extended to work for any group/set of subgroups?

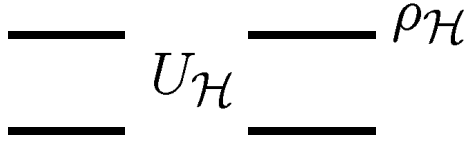
Open question 1: When are HSP and HSCP equivalent?

Why the HSCP?

The HSCP has more symmetry than the HSP

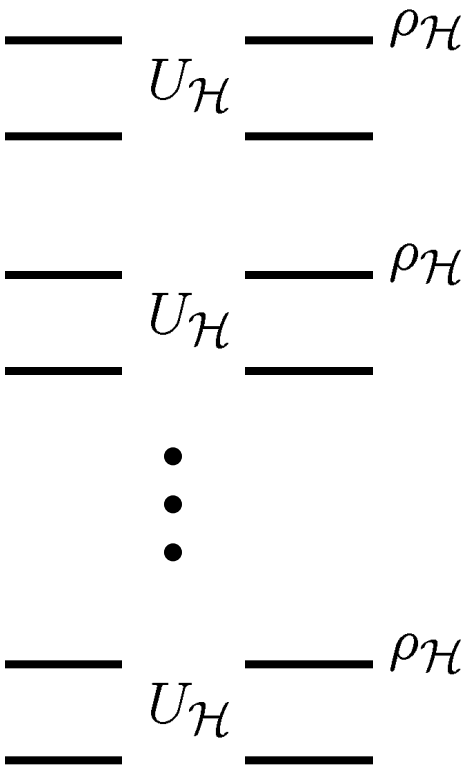


Multiple Queries



If we make a single measurement on the HS state, can we distinguish different $\mathcal{H}$

In general, NO

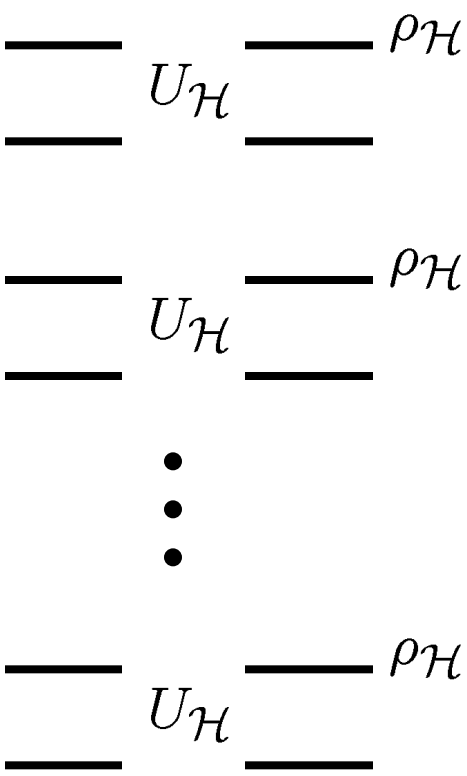


We need multiple queries to distinguish different subgroups

$\rho_{\mathcal{H}}^{\otimes m}$ $m = O(\log |\mathcal{G}|)$ sufficient

[Ettinger, Høyer, Knill 99]

Multiple Queries

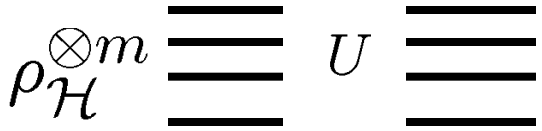


$\rho_{\mathcal{H}}^{\otimes m}$

Can we make a measurement on $m < O(\log |\mathcal{G}|)$ copies and learn the subgroup?

In general (and for important case of symmetric group) NO

[Moore, Russell, Hallgren, Roetteler, Sen 05]





Symmetries of HSCP States

Hidden subgroup conjugate state

$$\rho_{[\mathcal{H}]} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \rho_{g\mathcal{H}g^{-1}}$$

Hidden subgroup conjugate states

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\rho_{g\mathcal{H}g^{-1}})^{\otimes m}$$

In context of optimal measurement method (PGM)

[e.g. Ip 03, Bacon, Childs, and van Dam 05]

this translates into extra symmetry of the optimal measurement

$$\rho_{[\mathcal{H}]}^{\otimes m} = \bigoplus_{\mu_1, \ldots, \mu_m} \left(I_D \otimes \bigoplus_{\mu} \left(R_{[\mathcal{H}]} \otimes I_{d_\mu} \right) \right)$$

Symmetries of HSCP States

What is the extra symmetry of the multiple HSCP states?

Clebsch-Gordan transform for the regular representation of finite groups:

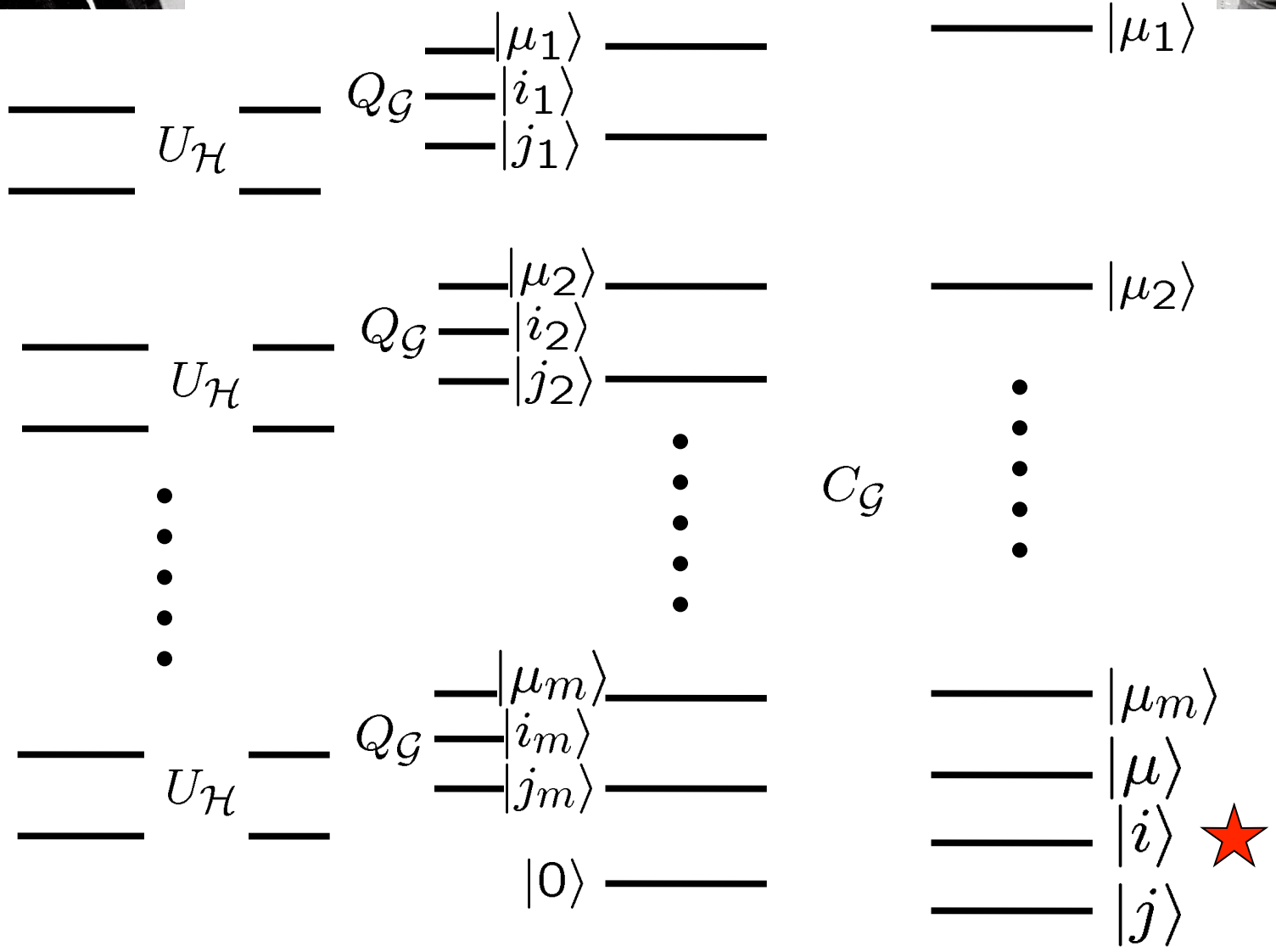
$$D_R(g)^{\otimes m} = \left[\bigoplus_{\mu}(I_{d_\mu} \otimes D_\mu(g))\right]^{\otimes m} \quad D_R(g)|g'\rangle = |g'g^{-1}\rangle$$

$$D_{\mu_1}(g) \otimes \cdots \otimes D_{\mu_m}(g) = \bigoplus_{\mu} \left(I_{n_\mu(\mu_1,\ldots,\mu_m)} \otimes D_\mu(g)\right)$$



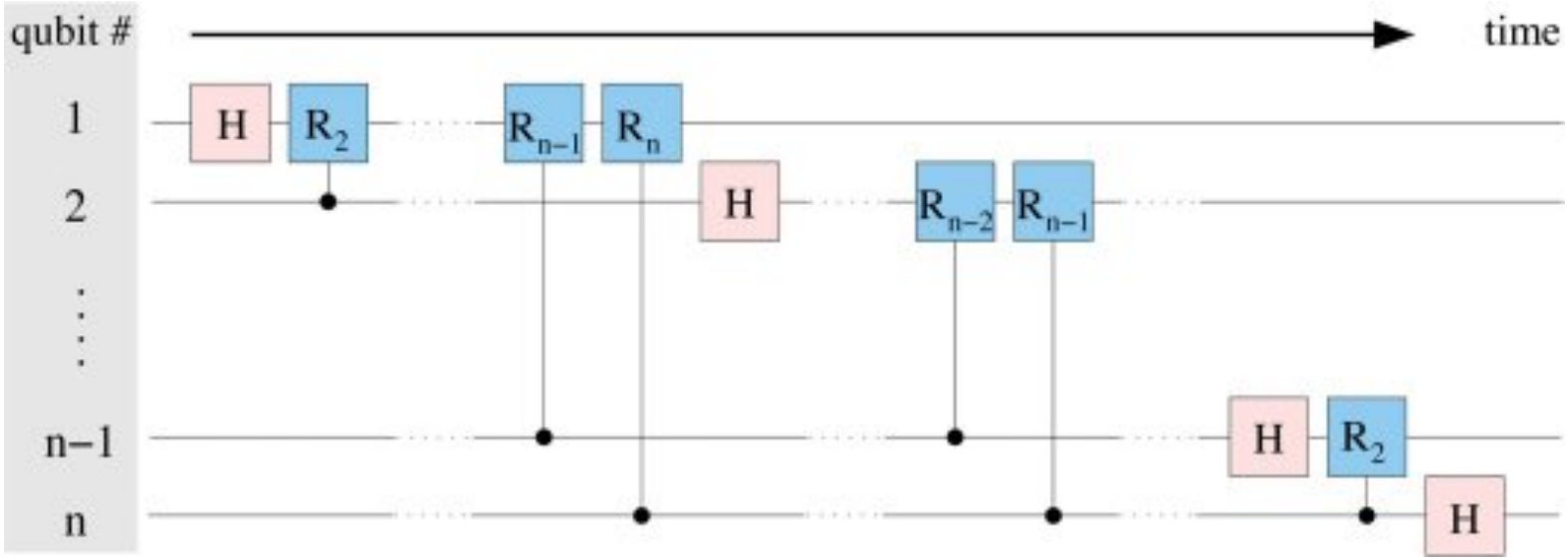
General Circuit





The Second Role of Symmetry

Without efficient circuits for the quantum Fourier transform, all is for naught!



Second role of symmetries:

Subgroup adapted basis provide a recursive structure useful for constructing representation theoretic transforms

[Moore, Rockmore, Russel 03] Generic QFTs

[Bacon, Chuang, Harrow 04,06] Schur Transform

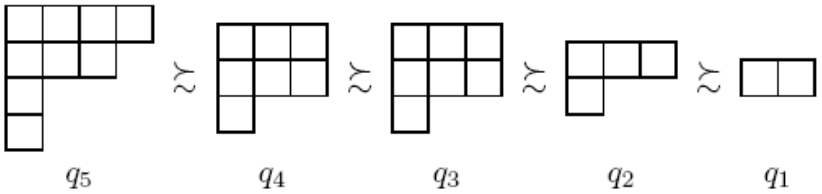
Clebsch-Gordan



[Bacon, Chuang, Harrow 04,06] Schur Transform

Clebsch-Gordan transform for U(d)

U(d) subgroup adapted basis (Gelfand-Zetlin basis)



Can these techniques be adapted for finite groups?

CG transform for U(d) is polynomial in d.

We want CG transform for regular representation of a finite group efficient in log of group size.

The To Do List!

Are the hidden subgroup conjugacy problem and the hidden subgroup problem equivalent?

For which groups are they equivalent?

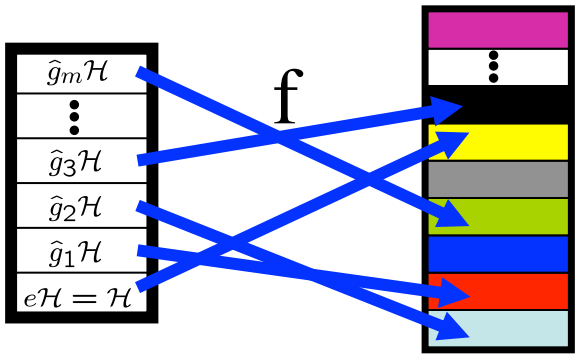
Explicit understanding of the Clebsch-Gordan symmetry involved in the dihedral and symmetric group hidden subgroup conjugacy problem.

Construction of efficient circuits for finite group Clebsch-Gordan transforms.

BIG QUESTION: Why are some non-abelian HSPs efficiently solvable? Is there a unifying picture?

The Hidden Subgroup Conjugacy Problem and Beyond

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Episode IV : A New Hope

